



Statement of Work	
Document Title: LCLS-II-HE EI B950 PCW System Upgrade & Room 310 CO2 Chillers Infrastructure Construction SOW	
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Document Approval:

Originator: Collin Chew, Senior Project Manager	<u>Collin Chew</u> Collin Chew (Mar 16, 2026 09:23:41 PDT)
• Content/Format • Content/ Format, Requirement ID's/Source ID's, Verification Planning, SME Input	

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• Facts, Details, Concepts, Calculations, Analysis to meet Stakeholders, Policies, Standards, Best Practices, as appl	
Reviewer: Canon Cheung, Project Manager – 1.06.04 Control Account Manager	<i>signature not required</i>
• Complete/Accurate • SME Involvement • Actionable, as appl	
Reviewer: Kathleen Ratcliffe – LINAC ES&H Manager	<i>signature not required</i>
• ES&H Content, as appl	
Reviewer: Scott DeBarger, Technical Director	<i>signature not required</i>
• Scope Alignment, Stakeholder/ SME Engagement	

Approver: Frank Bleisch– Supply Chain Director	
•General Procurement Provisions are Complete and Accurate	<u>Frank Bleisch</u> Frank Bleisch (Mar 16, 2026 11:06:03 PDT)
Approver: Quality Assurance Manager or designee	
• General QA Provisions • Req's Validation/Verification	<u>Rena George</u> Rena George (Mar 16, 2026 16:57:57 PDT)
Approver: Canon Cheung, 1.06 System Manager	
• Articulation, Meets Higher-Level Expectations, Stakeholder/SME Engagement, as appl	<u>Canon Cheung</u> Canon Cheung (Mar 16, 2026 09:27:32 PDT)

See [Document Review/Approval Matrix of Responsibilities](#)

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Revision Record

Revision Number	Revision Date	Description of Change
R3	3/16/2026	- Updated Section 7 – References to replace preliminary BIO design references with finalized construction drawing sets, specifications, and inspection records for the CO₂ Chiller Infrastructure, PCW Skid Demo & Installation, and PCW Branch work. - Expanded the reference list from 5 to 7 documents to support construction execution and inspection documentation.
R2	2/5/2026	- Removed section 5.2 Alternate Period of Performance due to clarity on Summer 2026 Shutdown Window and to optimize procurement workflow approval timeline - Updated section 5.1 with the following: Subcontractor acknowledges that portions of the work are strictly dependent on SLAC's Summer 2026 Shutdown Window, currently scheduled from July 20th, 2026, through August 31st, 2026
R1	1/26/2026	- Added language to clarify that certain construction activities are strictly dependent on SLAC's Summer 2026 Shutdown Window (July 20–August 20, 2026) and must be planned and executed within that period. - Added an optional Add Alternate Period of Performance allowing SLAC, at its discretion, to extend the shutdown window by two (2) weeks through August 31, 2026, with separate, itemized pricing. - Added clarification that the electrical distribution infrastructure is designed and sized to accommodate future expansion for up to six (6) CO₂ chiller skids. - Updated reference #2 and #3 with BIO PRS #26-009 and title update. - Updated reference #5 title update. - Added Figures 2-3, 2-4, and 2-5 to clarify project location and areas of work within Building 950A CUP – Room 900 and Building 950 NEH, including Room 213 Receiving Area and 1-Hall-W West Hallway.
R0	1/5/2025	Initial Release.

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1 Purpose

SLAC National Accelerator Laboratory (SLAC) is seeking a qualified and experienced General Contractor (Subcontractor) to provide construction services for the installation of the following infrastructure. This construction effort supports the LINAC Coherent Light Source – II – High Energy (LCLS-II-HE) Experimental Infrastructure (EI) project. Subcontractor shall furnish all labor, materials, equipment, tools, supervision, and services necessary to construct, install, and commission the mechanical, electrical, plumbing, and structural work identified in the design documents and specifications.

1. Room 310 CO2 Chiller Infrastructure
2. PCW Skid Demo and Relocation
3. PCW System Upgrade

Subcontractor shall coordinate closely with SLAC's Technical Systems and Facilities Engineering teams to ensure that adjacent systems remain operational, all shutdowns are properly scheduled, and that reusable components are preserved or staged per SLAC direction.

Room 310 CO2 Chillers Infrastructure

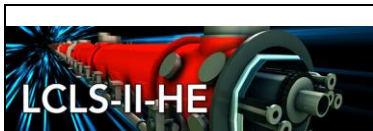
The purpose of the Room 310 CO2 Chillers Infrastructure scope is to construct and install the supporting mechanical, electrical, plumbing, and structural infrastructure required to accommodate two (2) CO₂ chiller skids that will be installed by a separate SLAC-managed contract. The electrical distribution infrastructure provided under this scope is designed and sized to accommodate future expansion for up to a total of six (6) CO₂ chiller skids. Subcontractor shall furnish all labor, materials, tools, equipment, supervision, and services necessary to complete the construction work as defined in the issued design documents and specifications.

PCW Skid Demo and Relocation

The purpose of the PCW Skid Demo and Relocation scope is to safely decommission, disconnect, and relocate the existing PCW skid and associated utilities to enable facility reconfiguration work under the Far Experimental Hall (FEH) Reconfiguration Project. Subcontractor shall provide all labor, materials, supervision, and equipment necessary to perform the removal, capping, labeling, and protection of affected systems in accordance with SLAC Design Standards, ESH requirements, and DOE safety policies.

PCW Skid Upgrade

The purpose of the PCW System Upgrade scope is to install a new PCW skid and upgrade the associated distribution systems in the basement and sub-basement of B950 to meet the increased cooling demand for the LCLS-II-HE program. The work includes the installation of the new PCW skid, upsizing of PCW supply and return lines, and implementation of temporary scaffolding and access modifications (including removal and reinstallation of limited ductwork and lighting) to perform the upgrades safely.



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2 Scope

Scope includes specific modification of the existing mechanical, plumbing, and electrical systems per the completed design drawings and specifications.

Mech Room 310 CO2 Chiller Infrastructure

The project is located within Building 950 NEH – Room 310. Room 310 is approximately 3059 square feet (sf). Figure 2-1 shows the Building 950 location. Figure 2-2 shows the Room 310's location.



Figure 2-1: Building 950 Location



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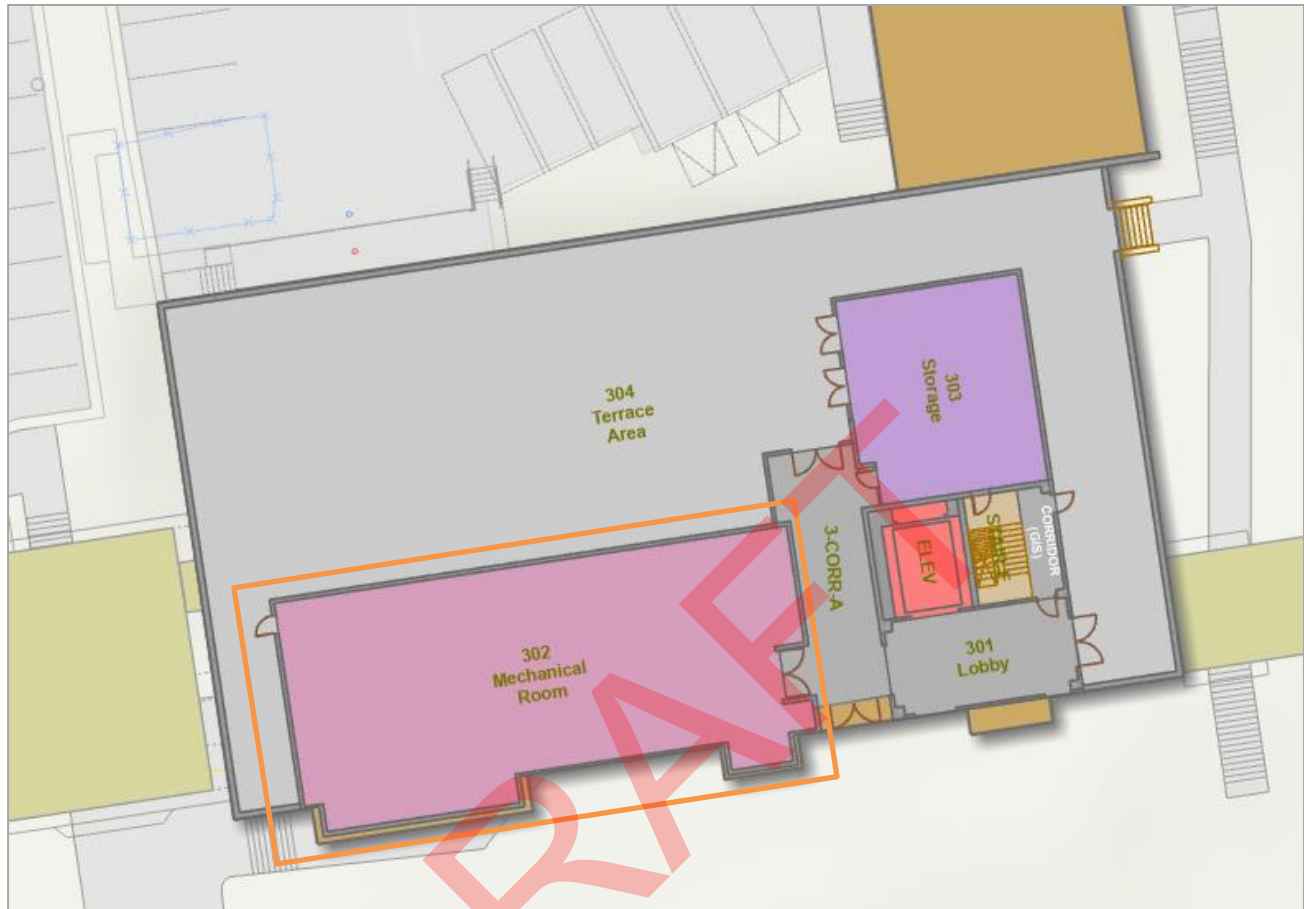


Figure 2-2: Building 950 Room 310 Location

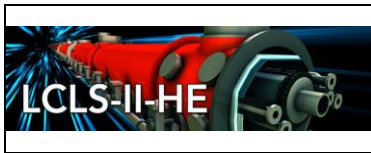
2.1 Room 310 CO2 Chiller Infrastructure

2.1.1 Subcontractor shall provide labor, shop drawings, materials, and resources required to complete structural and architectural modifications, mechanical, plumbing, and electrical systems, and necessary demolitions according to the construction authorization and inspection record requirements, construction drawings, and specifications reference #1, #2 and #3. This includes:

- a) Installation of mechanical piping, valves, and associated utilities necessary to support the chiller operation.
- b) Installation of electrical distribution infrastructure, conduit, and wiring to provide power and controls for future SLAC provided CO₂ skids.
- c) Architectural and facility modifications within Room 310 to support equipment access, installation clearances, and safety requirements.
- d) Transportation and relocation of the 4ATS1A-950 transfer switch to the designated new location as identified in the construction drawings supporting the CO₂ Chiller Infrastructure Project.

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- e) Coordination with SLAC technical teams and other contractors to ensure seamless interface with the separately managed future SLAC CO₂ skid installation and system commissioning.
- 2.1.2 Subcontractor shall coordinate utility location/scanning of slab prior to drilling. All penetrations and their shielding shall be in accordance with Radiation Physics' requirements.
- 2.1.3 Subcontractor shall coordinate with SLAC PM and CM for site access for shop drawing development. Subcontractor shall coordinate with SLAC CM to complete the required Work Planning and Control (WPC) documents and obtain release to execute construction tasks. Subcontractor shall ensure best and safe practices are followed and shall protect existing equipment as required.
- 2.1.4 Subcontractor shall coordinate utility location/scanning of slab prior to drilling. All penetrations and their shielding shall be in accordance with Radiation Physics' requirements.
- 2.1.5 Subcontractor shall coordinate with SLAC PM and CM for site access for shop drawing development. Subcontractor shall coordinate with SLAC CM to complete the required Work Planning and Control (WPC) documents and obtain release to execute construction tasks. Subcontractor shall ensure best and safe practices are followed and shall protect existing equipment as required.



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2.2 B950A PCW Skid Demo and Relocation

The project is located within Building 950A Central Utilities Plant (CUP) – Room 900. Room 900 is approximately 1743 square feet (sf). Figure 2-3 shows Room 900's location.

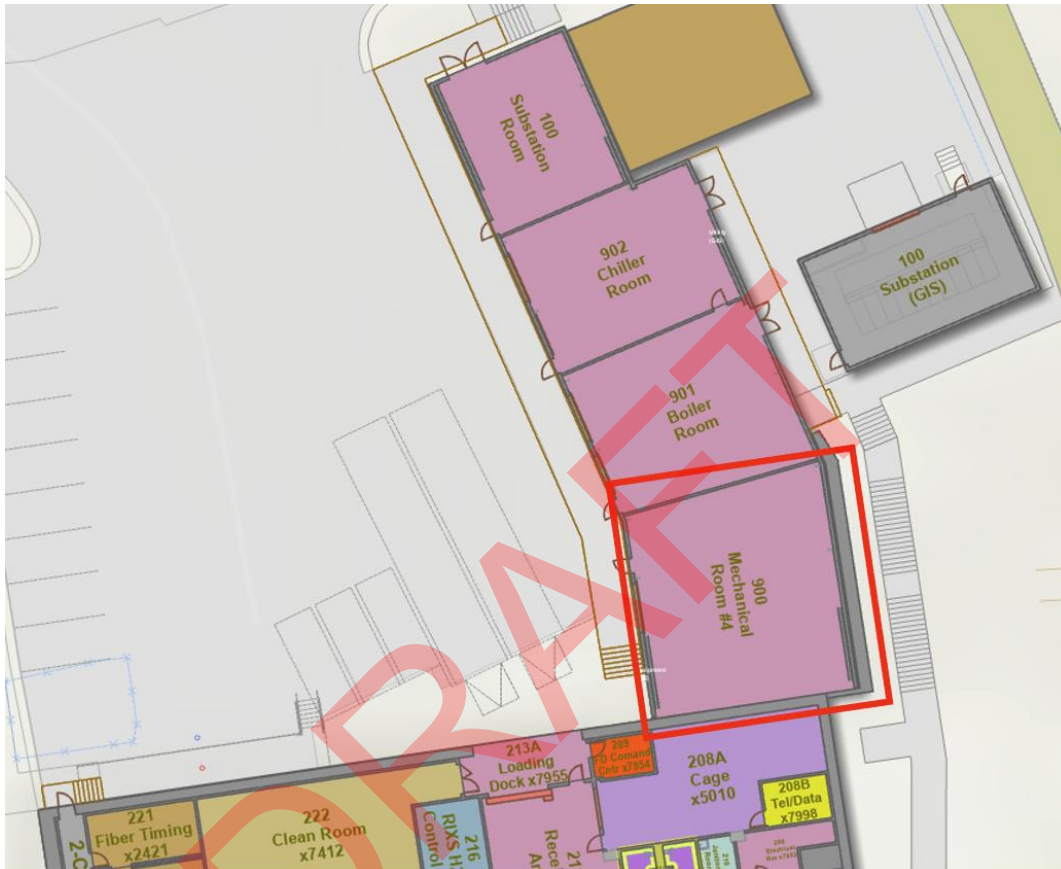


Figure 2-3: Building 950A Room 900 Location

- 2.2.1 Subcontractor shall provide labor, shop drawings, materials, and resources required to complete structural and architectural modifications, mechanical, plumbing, and electrical systems, and necessary demolitions according to the construction authorization and inspection record requirements, construction drawings, and specifications reference #4, #6, and #7. This includes:
- Disconnect and safely remove the existing B950A PCW skid located in the NEH, including associated mechanical, electrical, and control connections.
 - Perform necessary demolition of piping, supports, and ancillary equipment for relocation, including flushing.
 - Relocate reusable components and equipment to the designated staging or storage area identified by SLAC.
 - Cap and label all disconnected utilities per SLAC standards.

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- e) Coordinate with SLAC's Technical Systems and Facilities Engineering to ensure that no disruption occurs to other critical cooling loops.
- f) Manage and dispose of all demolition debris and fluids per SLAC ESH and waste management procedures.

2.3 B950 PCW System Upgrade

The project is located within the following areas:

Building 950A Central Utilities Plant (CUP) – Room 900.

Room 900 is approximately 1743 square feet (sf). Figure 2-3 in the previous section shows Room 900's location.

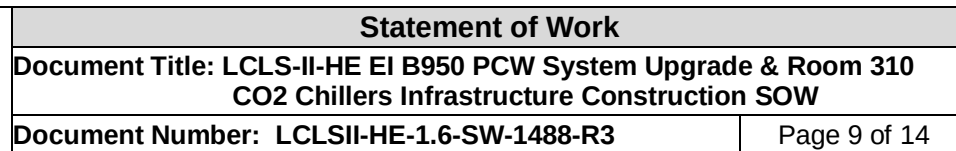
Building 950 NEH – Room 213 Receiving Area and 1-Hall-W West Hallway

The Room 213 Receiving Area is approximately 693 square feet (sf). Figure 2-4 shows the Room 213 location.

The 1-Hall-W West Hallway is approximately 2440 square feet (sf). Figure 2-5 shows the 1-Hall-W West Hallway location.



Figure 2-4: Building 950 Room 213 Receiving Area Location



2.3.1 Installation of New PCW Skid

- a) Receive, offload, and install the new PCW skid at the designated NEH location.
- b) Connect to existing PCW supply and return headers, electrical power, and controls per the design documents.
- c) Perform alignment, anchorage, vibration isolation, and torque verification per manufacturer specifications.
- d) Support start-up, flushing, and commissioning of the new skid in coordination with SLAC Technical Systems and Commissioning teams.
- e) Provide as-built documentation, O&M manuals, and test reports upon completion.

- a) Remove existing PCW piping sections and install new, larger-diameter piping as indicated in the IFC drawings.
- b) Provide temporary scaffolding and access platforms to safely perform work in the sub-basement.

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- c) Remove and later reinstall a portion of existing ductwork and one (1) light fixture to gain access to the PCW lines.
- d) Pressure test, flush, and label new piping per SLAC mechanical standards.
- e) Coordinate with SLAC CM to sequence work to minimize disruption to nearby systems.

2.4 Hydronic Balancing Requirements

Subcontractor shall provide all labor, materials, instrumentation, and technical expertise necessary to perform **hydronic balancing (TAB)** for all process cooling water (PCW) and chilled-water systems affected by this project. Balancing shall be included for the following scopes:

2.4.1 PCW System Upgrade – Verify and balance design flows resulting from modifications to pumps, distribution headers, control valves, and associated piping. Subcontractor shall confirm that all connected loads receive the specified flow rates and that the upgraded skid operates within design pressure and performance parameters.

2.4.2 PCW Skid Demolition and Relocation – Following removal and reinstallation of the skid, subcontractor shall balance the system to account for changes in piping length, elevation, pressure losses, and branch distribution. Recommissioning shall include verification of pump performance, valve settings, and flow stability throughout the PCW network.

2.4.3 Room 310 Chiller Infrastructure – Subcontractor shall perform hydronic balancing for all new or modified chilled-water piping, coils, control valves, and equipment introduced under this project. If tie-ins modify existing flow distribution, balancing shall extend to upstream or downstream branches as required to restore system performance to design conditions.

2.5 Meetings

1. Along with the Subcontractor Safety and Quality Assurance representatives, Subcontractor shall attend morning safety tailgate meetings for each construction workday.
2. Subcontractor shall submit a daily report to the SLAC Construction Manager after the end of the work shift.

2.6 Inspections, Testing, Acceptance

1. Subcontractor shall submit QA & QC plans and documentation for SLAC review at the KOM and designate a QA Representative. Subcontractor shall adhere to approved QA & QC plans throughout the execution of the project.
2. Subcontractor shall complete code required testing for modified and new installations, as specified in the drawing package.
3. Subcontractor shall submit test plans and procedures for SLAC review and approval prior to testing the system. Subcontractor shall submit completed test reports to SLAC.
4. Subcontractor shall coordinate with SLAC PM and CM as work nears completion to schedule SLAC testing and acceptance.

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5. Subcontractor shall complete any punch list items and ensure results are satisfactory prior to scheduling a follow-up test and inspection.

2.7 Project Closeout

1. Subcontractor shall submit construction as-built drawings and details in pdf format, Operations and Manuals (O&Ms), cutsheets, and documents related to the completed scope.
2. Subcontractor shall maintain a clean site through construction, clean the work areas prior to site demobilization. Waste disposal shall be coordinated with the SLAC CM.

2.8 Project Expectations

1. SLAC will coordinate the removal of applicable existing technical equipment within the scope of work prior to Subcontractor mobilization.
2. Subcontractor shall provide all labor, materials, tools, equipment, and services needed to perform the construction work described in this scope, drawings, and specifications, or that may be reasonably inferred as being necessary for the completion and serviceability for the intended use of all items of work.
3. Subcontractor shall complete the required SLAC training courses to be able to access the site and perform the scope of work. The training will consist of both virtual and on-site practical courses. These courses shall be completed prior to executing the field modifications.
4. Once drawings and materials are approved by SLAC, Subcontractor shall not change nor substitute these without the explicit approval of SLAC Project Management.
5. Subcontractor shall comply with the following requirements:
 - a. Construction Limits and Temporary Facilities
 - i. Location and installation of temporary facilities shall be approved by SLAC through the Building Inspection Office (BIO) review process. Existing driveways, traffic and fire lanes affected by construction shall always remain open during construction. On-site fire hydrants and Building perimeter areas shall be accessible by fire trucks as required by the SLAC Fire Marshal's Office.
 - b. Utility Systems:
 - i. Temporary Construction Power to the construction site shall be coordinated with the SLAC CM.
 - c. Environmental, Safety, Health (ESH) Criteria:
 - i. Subcontractor shall incorporate into the facilities design, consideration for the safety and health of the employees and the public and mitigate impacts to the environment. This shall take precedence over other aspects, such as function, cost, and expediency. Subcontractor shall adhere and comply to applicable sections of the SLAC ES&H Manual, and obtain guidance from the SLAC CM.
6. SLAC Provided Equipment, Services, and Information
 - a. SLAC will provide all new lighting fixtures for subcontractor to install.
 - b. SLAC will provide waste bins and drums. Subcontractor shall inform SLAC CM 48 hours prior for delivery.

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- c. SLAC will provide electrical power for construction activities.

3 Overview of Safety Considerations and Requirements

The following outlines SLAC's approach to the flow down of Safety requirements that when fully utilized will reduce the risk of accident and injury. Subcontractor shall adhere to SLAC's ES&H program which is encapsulated in Section 9.1 below. Refer to the Specifications for additional requirements.

- 3.1 The SLAC Environmental, Safety, and Health Division (ESH) manual Chapter 42 Subcontractor Safety provides the policies and procedures by which SLAC meets its 10 CFR 851 & Cal/OSHA requirements. Integrated Safety and Environmental Management System (ISEMS) is used to systematically integrate safety and environmental management into the execution of any project at SLAC. Ensuring adequate protection of the public, the workers and the environment is integral to the design of the Project. See <https://www-group.slac.stanford.edu/esh/general/isems/> for additional information.
- 3.2 Installation plans for temporary staging facilities, including portable toilets, and fencing shall be provided by Subcontractor.
- 3.3 Subcontractor shall implement a Construction Readiness Review (CRR) with all key personnel from subcontractor and main lower sub-tiers prior to start the construction. The schedule of this review shall be coordinated with SLAC Procurement Officer and Project Manager or his designee.
- 3.4 Use of trained and qualified employees
- 3.5 Subcontractor shall only use employees who are trained and qualified for their assigned work.

4 Project Deliverables

Subcontractor shall deliver the following to SLAC:

- 4.1 An upgraded high quality mechanical and electrical system in Building 950 Room 310 to support the installation of two (2) future SLAC provided CO2 Chiller Skids managed by a different team.
- 4.2 The safe removal, disconnection, and relocation of the existing B950A PCW skid located in the Near Experimental Hall (NEH), including capping, labeling, and protection of utilities to support FEH Reconfiguration work.
- 4.3 A new PCW skid installation with associated piping upgrades in the basement and sub-basement, including complete restoration of all affected areas upon project completion.
- 4.4 Required documents:
 - 4.4.1 Personnel Roles and Responsibilities
 - 4.4.2 Project Schedule
 - 4.4.3 QA/QC Plan

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4.4.4 Shop Drawings for BIO Approval

4.4.5 Work Planning and Control Documents

- a. Job Safety/Hazard Analysis
- b. Lock-Out/Tag-Out (LOTO) Plan

5 Period of Performance and Project Milestones

Subcontractor shall provide a project schedule that takes into consideration and meets duration and dates in Table 1. The schedule will be reviewed on a weekly basis to identify overall progress and highlight any issues or concerns. Subcontractor shall also provide a four-week look ahead schedule to identify critical tasks that need coordination to be reviewed each week.

5.1 Base Period of Performance

Subcontractor acknowledges that portions of the work are strictly dependent on SLAC's Summer 2026 Shutdown Window, currently scheduled from July 20th, 2026, through August 31st, 2026. All Activities requiring site access, coordination with shutdown conditions, or integrations with shutdown-dependent work shall be planned, supported, and completed by Subcontractor within this window.

Table 1: Period of Performance and Project Milestones

Milestone	Details	Weeks after NTP
0	Subcontract Award	N/A
1	Notice to Proceed	0
2	Work Planning Control Complete	2 weeks
3	Shop Drawing and Submittal Complete	2 weeks
4	SLAC BIO Approved Submittal	3 weeks
6	Construction Complete	9 weeks
9	Testing, Inspections and Acceptance Complete	10 weeks
10	Site Clean-up and Project Closeout	10 weeks

6 Definitions

Term	Definition
PM	Project Manager
CM	Construction Manager
BIO	Building Inspection Office
MEP	Mechanical, Electrical, Plumbing

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WPC	Work Planning Control
NEH	Near Experimental Hall

7 References

Reference Number	Document Title
Reference #1	Inspection Record - CO2 Chiller Infrastructure
Reference #2	CO2 Chillers Infrastructure - Drawing Set
Reference #3	CO2 Chillers Infrastructure - Specifications
Reference #4	PCW Skid Demo & Installation - Drawing Set
Reference #5	XPP Design PCW Branches
Reference #6	XPP Design Specifications
Reference #7	Inspection Record - PCW Skid

8 Applicable Design Standards, Codes, And Guidelines

Refer to document LCLSII-HE-1.6-TS-0298 and technical specification for all the design standards, codes, and guidelines.